# PhD Math Camp

## Calculus II: Integrals

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# Introduction

- In this lecture we will learn what **integrals** are and how are they useful.
- But we will motivate integrals from a specific angle: **probability**.
  - We will learn that integrals help compute probabilities for continuous variables.
  - Also, for continuous variables, integrals define important properties (aka, *parameters*). For example, the true mean of a population is defined with an integral.
- Integrals show up in many applications, but fortunately computers will do the calculations. *We just need to know what an integral is, and what does it mean when it shows up in a specific application*.

# Sample Space, Events, Random Variables, and Probability

- Sample space, $\Omega$, includes all possible outcomes of a random process.
- An event is a subset of the sample space, e.g., all elements $x_i \in \Omega$ where a property $Q(x_i)$ is true.
- A random variable (RV), $X$, is numerical and represents a collection of outcomes of a random process (an event). $X$ can be discrete (integer subset) or continuous (real subset).
- The probability of an event is the count of favorable outcomes (i.e., the event happens) divided by all outcomes.
- Generally, for any discrete RV, $X$, then

$$P(X = x_i) = \frac{|\{\omega_j \in \Omega : \omega_j \leftrightarrow X = x_i\}|}{|\Omega|} = \frac{\#\text{Favorable Outcomes}}{\#\text{All Outcomes}}$$

# Discrete vs. Continuous Random Variables

- Discrete Random Variable (DRV):
  - Let $X$ be a DRV, then $X$ takes on values from a subset of integers, Z.
  - Probability function is defined as, $f : X \to [0, 1]$, such that for each $x_i \in X$, the probability function $f(x_i)$ gives the probability that $X$ equals $x_i$.
- Continuous Random Variable (CRV):
  - Let $X$ be a CRV, then $X$ takes on infinitely many values from a real subset.
  - Probability function is defined as, $f : C \subseteq X \to [0, 1]$, such that for any subset $C = [a, b] = \{x_i \in X : x_i \in [a, b]\}$, the probability function $f(C)$ yields the probability that $X$ lies in $C$.

# From the Probability Distribution Function to the Cumulative Distribution Function (CDF)

**The CDF:**

- The Cumulative Distribution Function (CDF) calculates the probability of obs. $x_i$ or a lower value. It sums all probabilities up to $x_i$. For a Discrete RV:

$$\Pr(X \leq x_i) = CDF(x_i) = F(x_i) = \sum_{k=-\infty}^{i} f(x_k)$$

- Thus, starting from a Probability Distribution Function, we can always construct the Cumulative Distribution Function by adding up the probabilities from the minimum possible value of $X$ up to the probability of $x_i$

# From the Cumulative Distribution Function (CDF) to the Probability Distribution Function

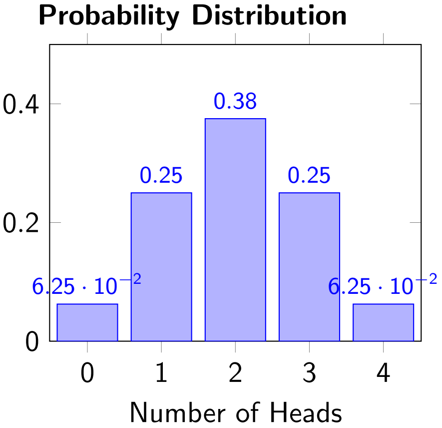
**Deriving Probability from CDF:**

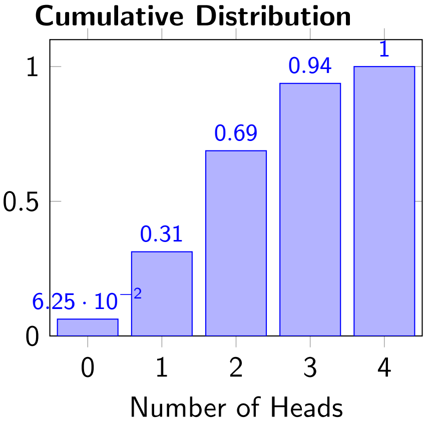
- With the CDF of a random variable, you can find its Probability Distribution Function.
- For a Discrete RV, simply subtract the CDF values at two successive points to find the probability of the latter point:

$$F(x_i) - F(x_{i-1}) = \sum_{k=-\infty}^{i} f(x_k) - \sum_{k=-\infty}^{i-1} f(x_k) = f(x_i)$$

- This allows you to calculate probability at a point from accumulated probability.

# Probability Distribution and Cumulative Distribution of the Total Number of Heads When Throwing Four Coins



# Key Takeaway: Interplay of the Probability Distribution and the CDF

- When you look at a discrete Probability Distribution Function and CDF, the Probability Distribution at $x = a + 1$ equals exactly the rate of change in accumulated probability when one moves from $a$ to $a + 1$ in the CDF.
- Metaphorically, the Probability Distribution seems to represent the “speed” at which we accumulate probability.
- Similarly, the CDF value at a given point $a$, is the sum of all probability “speeds” from the minimum possible value of $X$ to $a$.
- This is a nice connection: **summing the areas under the Probability Distribution graph yields the CDF, and the “slope” of the CDF graph recovers the Probability Distribution**.

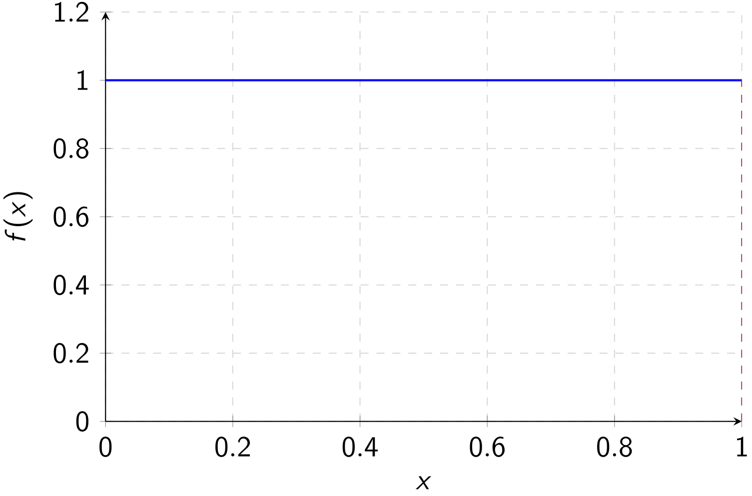
# Continuous Random Variables and the Probability Density Function (PDF)

- *But*, continous random variables have an important difference – **the probabilities are no longer points, but areas under a curve**.
- An infinitely thin slice in continuous space contributes no area, so to compute probabilities we use closed intervals, $[a, b]$, and a function, $f(x_i)$ that compues the *relative likelihood* of $x_i \in [a, b] \subseteq X$.
- We call $f$, the Probability **Density** Function (PDF). If $f$ is the PDF of a Continuous RV $X$, then $f$ needs to satisfy two conditions:
  1. The PDF is always non-negative, i.e. $f(x) \geq 0$ for all $x$ in the domain of $X$.
  2. The area under the entire PDF curve is 1.

# Uniform Distribution for Continuous Random Variables

- Consider the uniform distribution on the interval [0, 1].
- The PDF is:
$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$
- The total area under the PDF curve is 1.

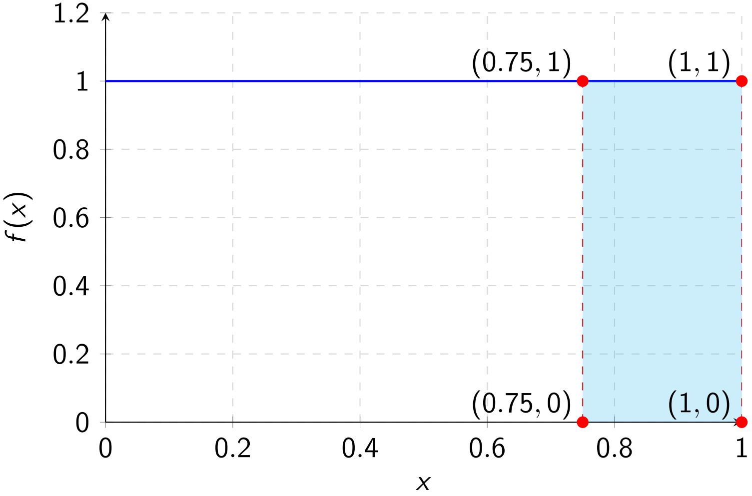
# Graph of the Uniform Distribution PDF



# Probability Calculation for Uniform Distribution

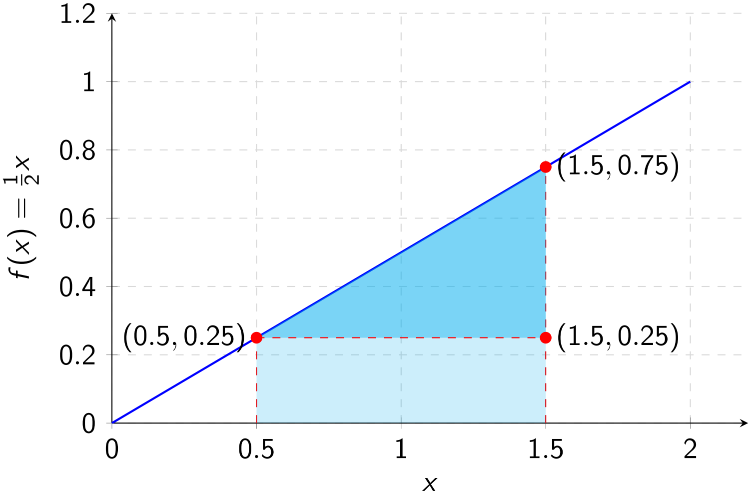
- The probability that $X$ is greater than 0.75 in a uniform distribution from 0 to 1 is calculated as follows:

$$P(0.75 < X) = 1 \times (1 - 0.75) = 0.25$$



# Another Example: a Triangular PDF

- Suppose $f(x_i) = (1/2)x_i \leftrightarrow x_i \in [0,2]$ and $f(x_i) = 0 \leftrightarrow x_i \notin [0,2]$.
- To calculate the probability that $X$ lies between 0.5 and 1.5 we can use one rectangle and one triangle to represent the area.



# How do we compute areas under a curve?

- So, given the previous definition, for any Continous RV $X$ with PDF $f(x_i)$, to calculate the probability that $X$ lies in some interval $[a, b]$ we need to compute the area under the PDF curve between $a$ and $b$.
- How do we do that for *any* curve $f(x_i)$?

# Approximating Area: Riemann Sums

- **Riemann sums** are a method to approximate the area under a curve by breaking it down into simpler geometric shapes, such as rectangles.
- To compute a Riemann sum, we:
  1. Divide the interval $[a, b]$ into $n$ subintervals of equal width, $\Delta x = \frac{b-a}{n}$.
  2. On each subinterval, construct a rectangle. The height of each rectangle is determined by the function value at a specific point within the subinterval.
  3. Sum the areas of all rectangles to get an approximation of the total area under the curve.

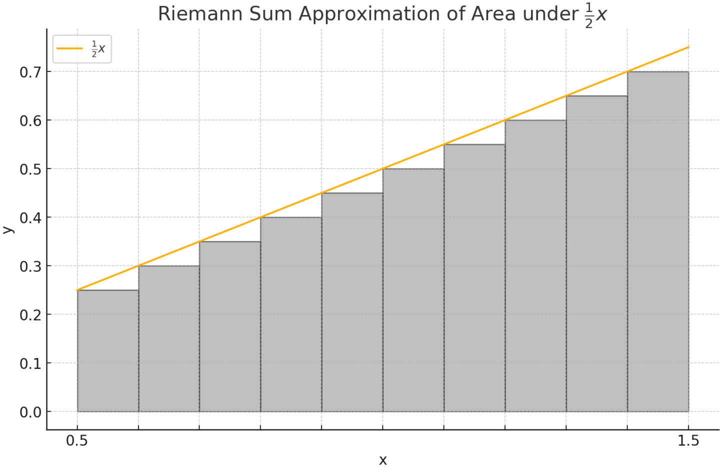
# Approximating Area: Riemann Sums

- The Riemann sum can be written as:

$$S_n = \sum_{i=1}^{n} f(x_i^*) \cdot \frac{b-a}{n} = \sum_{i=1}^{n} f(x_i^*)\Delta x$$

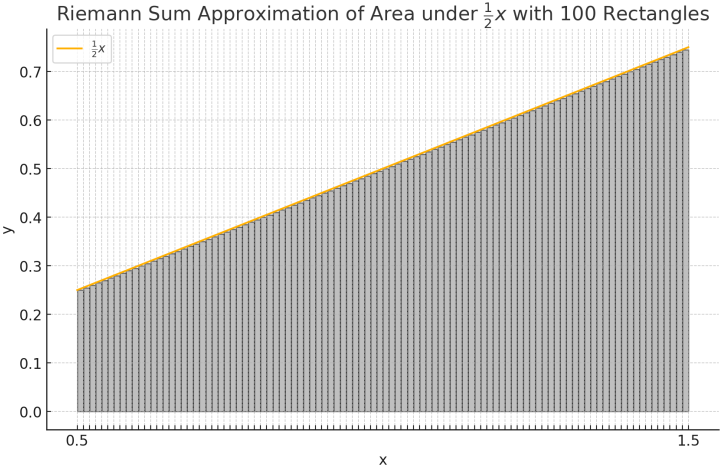
  where $x_i^*$ is a chosen point in the $i$-th subinterval.
- As $n$ increases (i.e., more rectangles), the approximation becomes more accurate, and in the limit as $n \to \infty$, the Riemann sum converges to the exact area, which is the integral of the function.

# Triangular PDF: Riemann Approximation with 10 Rectangles




- Riemann Sum = 0.475. Exact Value = 0.50.

# Triangular PDF: Riemann Approximation with 100 Rectangles




- Riemann Sum $= 0.4975$. Exact Value $= 0.50$.

# The Riemann Integral: Taking the Limit

- While Riemann sums provide an approximation, the **Riemann integral** gives the exact area under the curve.
- The Riemann integral is defined as the limit of the Riemann sum as the number of subintervals $n$ approaches infinity:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x = \lim_{||\Delta x||\to 0} \sum_{i=1}^{n} f(x_i^*)\Delta x_i$$

- Here:
  - $\Delta x = \frac{b-a}{n}$ is the width of each subinterval.
  - $x_i^*$ is a chosen point in the $i$-th subinterval $[x_{i-1}, x_i]$.
  - The sum $\sum_{i=1}^{n} f(x_i^*)\Delta x$ represents the total area of the rectangles approximating the area under $f(x)$.
- The integral $\int_a^b f(x)\,dx$ represents the exact area

# Key Takeaway: Riemann Integrals and Their Challenges

- **Riemann integrals** provide a clear and intuitive way to understand the area under a curve by summing up the areas of rectangles and taking the limit as the number of rectangles approaches infinity.
- *Intuitive interpretation*: The integral represents the exact area under the curve $f(x)$ between two points $a$ and $b$.
- **However**, calculating the limit of these sums to find the exact area can be tedious and complex, especially for complicated functions.
- This complexity raises the question: Is there a more efficient way to compute these areas without directly taking limits?

# Introduction to Antiderivatives

- An **antiderivative** of a function $f(x)$ is another function $F(x)$ whose derivative is $f(x)$.
- In other words, $F(x)$ satisfies:

$$F'(x) = f(x)$$

- *Example*: If $f(x) = 2x$, then one possible antiderivative is $F(x) = x^2$ because:

$$\frac{d}{dx}\left(x^2\right) = 2x$$

- **Key Idea**: The process of finding an antiderivative is often called *integration*, and it is the reverse of differentiation.

# Antiderivatives and the Constant $C$

- When we find the antiderivative (indefinite integral) of a function $f(x)$, we obtain a **family of functions** rather than a single function.
- The general form of an antiderivative is given by:

  $$F(x) + C$$

  where $C$ is an arbitrary constant.
- **Key Insight:** The constant $C$ accounts for the fact that differentiation of a constant yields zero, meaning any function of the form $F(x) + C$ has the same derivative $f(x)$.
- *Example*: For $f(x) = 2x$, the antiderivative is:

  $$\int 2x \, dx = x^2 + C$$

  where $C$ could be any real number.

# Common Rules for Finding Indefinite Integrals

| Rule | Indefinite Integral |
|---|---|
| **Power Rule** | $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$ |
| **Constant Multiple Rule** | $\int k \cdot f(x) \, dx = k \cdot \int f(x) \, dx$ |
| **Sum/Difference Rule** | $\int (f(x) \pm g(x)) \, dx = \int f(x) \, dx \pm \int g(x) \, dx$ |
| **Exponential Rule** | $\int e^{kx} \, dx = \frac{1}{k} e^{kx} + C$ |
| **Trigonometric Integrals** | $\int \sin(x) \, dx = -\cos(x) + C$<br>$\int \cos(x) \, dx = \sin(x) + C$<br>$\int \sec^2(x) \, dx = \tan(x) + C$ |
| **Logarithmic Rule** | $\int \frac{1}{x} \, dx = \ln \lvert x \rvert + C \quad (x \neq 0)$ |

# The Fundamental Theorem of Calculus: Part 1

- The Fundamental Theorem of Calculus (FTC) connects differentiation with integration, offering a practical way to compute integrals.
- **FTC Part 1:** If $F(x)$ is an antiderivative of $f(x)$ on an interval $[a, b]$, then:

$$\frac{d}{dx}(F(x)) = \frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

- This means that the derivative of the integral function of $f(x)$ (from $a$ to $x$) is equal to the original function $f(x)$.
- **Key Insight:** Differentiation and integration are inverse processes.

# The Fundamental Theorem of Calculus: Part 2

- **FTC Part 2:** If $F(x)$ is any antiderivative of $f(x)$, then the definite integral of $f(x)$ from $a$ to $b$ is:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

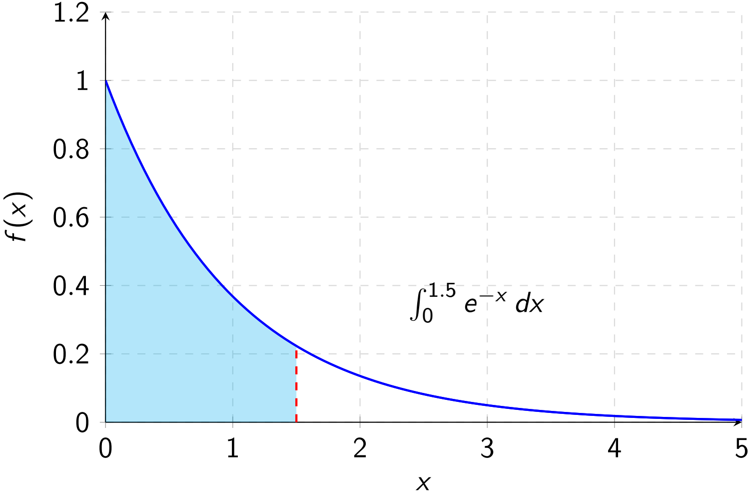
- **Key Insight:** Instead of using Riemann sums to find the area under $f(x)$, we can find an antiderivative $F(x)$ and evaluate it at the endpoints $a$ and $b$.
- *Example*: If $f(x) = 2x$, then:

$$\int_0^2 2x\,dx = \left[x^2\right]_0^2 = F(2) - F(0) = 2^2 - 0^2 = 4$$

# Example: Exponential Distribution

- Consider the exponential distribution with rate parameter $\lambda = 1$:

$$f(x) = e^{-x} \quad \text{for } x \geq 0$$



# Example: Computing Probabilities Using the Exponential Distribution

- Consider a random variable $X$ that follows an **exponential distribution** with rate parameter $\lambda > 0$. The PDF is:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

- **Question:** What is the probability that $X$ is less than or equal to a value $a$?

$$P(X \leq a) = CDF(X) = \int_0^a \lambda e^{-\lambda x} \, dx$$

- **Solution:**

$$\int_0^a \lambda e^{-\lambda x} \, dx = \left[ -e^{-\lambda x} \right]_0^a = 1 - e^{-\lambda a}$$

- Thus, $P(X \leq a) = 1 - e^{-\lambda a}$.

# The Normal Distribution

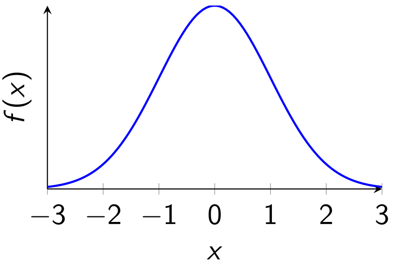
- **Normal Distribution**:
  - Continuous probability distribution defined by mean $\mu$ and standard deviation $\sigma$.
  - PDF:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **Computing Probabilities**:
  - Probability of an interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

  - CDF: $F(x) = \int_{-\infty}^{x} f(t)\,dt$



# Standard Normal Distribution

- **Standard Normal Distribution**:
  - A special case of the normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$.
  - The probability density function (PDF) is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

- **Computing Probabilities using Integrals**:
  - For $a \leq Z \leq b$, the probability is:

$$P(a \leq Z \leq b) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx$$

  - The CDF denoted by $\Phi(x)$, is expressed as:

$$\Phi(x) = P(Z \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\, dt$$

# Computing Probabilities for the Standard Normal Distribution

The area under the PDF from $-\infty$ up to $x$ defines the $CDF(x) = P(Z \leq x)$.

- Interactive Demo

# Derivatives, Integrals, and Probability

- **Calculating Changes Over Intervals**:
  - To find the change in any $F(x)$ from $x = a$ to $x = b$, just compute $F(b) - F(a)$.
  - If $F(x)$ is the CDF, then $F(b) - F(a) = \text{CDF}(b) - \text{CDF}(a) = P(a < x < b)$.
- **PDF as a Derivative**:
  - What if you only have the PDF $f(x)$? Here's a twist: think of the PDF as the derivative of the CDF. Yep, that's right—PDF is the rate of change of probability for a tiny change in $x$.
- **Summing It All Up: The Integral**:
  - To find $P(a < x < b)$, sum up all those tiny changes in probability (that is, integrate the PDF):

$$P(a < x < b) = \int_a^b f(x)\, dx = F(b) - F(a).$$

  - Voilà! Integrals save the day by summing infinite tiny contributions!

# Expectation as the True Mean: The Center of Mass

- **True Mean (Expectation) of $X$:**
  - The expectation $\mathbb{E}[X]$ represents the true mean of a random variable $X$.
  - It is defined as:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

    where $f(x)$ is the probability density function (PDF) of $X$.
- **Intuition: Center of Mass**
  - Think of the expectation as the "center of mass" of the probability distribution.
  - Just as the center of mass is the point where the mass of a physical object is balanced, $\mathbb{E}[X]$ is the point where the distribution of probabilities balances out.
  - This center balances all the possible values of $X$, weighted by their relative likelihoods (given by $f(x)$).

# Numerical Integration: The Basics

- **Purpose:**
  - Approximate the integral $\int_a^b f(x)\,dx$ when an analytical solution is difficult or impossible.
- **Key Concept:**
  - Break the interval $[a, b]$ into $n$ small subintervals of equal width $\Delta x = \frac{b-a}{n}$.
  - Approximate the area under the curve $f(x)$ on each subinterval.
- **Basic Approach:**
  - The integral is approximated by summing up these small areas:

$$\int_a^b f(x)\,dx \approx \sum_{i=1}^{n} f(x_i) \cdot \Delta x$$

  - $x_i$ represents the chosen point in the $i$-th subinterval.

# Example: Trapezoidal Rule for $\int_0^1 e^{-x^2}\,dx$

- **Goal:** Approximate $\int_0^1 e^{-x^2}\,dx$ using the Trapezoidal Rule with $n = 4$ subintervals.
- **Step 1: Subinterval Width:** $\Delta x = \frac{1-0}{4} = 0.25$.
- **Step 2: Function Values:** $f(0) = 1,\ f(0.25) \approx 0.939,\ f(0.5) \approx 0.778,\ f(0.75) \approx 0.569,\ f(1) \approx 0.368$
- **Step 3: Apply Trapezoidal Rule:**

$$\int_0^1 e^{-x^2}\,dx \approx \frac{0.25}{2}\left[1 + 2(0.939 + 0.778 + 0.569) + 0.368\right]$$

$$\int_0^1 e^{-x^2}\,dx \approx 0.743$$

# Monte Carlo Integration: Step by Step

- **Step 1: Define the Integral**. Let $\int_a^b f(x)\,dx$ be the integral you want to approximate.
- **Step 2: Generate Random Samples**. Generate $N$ random points $x_1, x_2, \ldots, x_N$ uniformly in the interval $[a, b]$.
- **Step 3: Evaluate the Function**. Compute $f(x_i)$ for each random sample $x_i$.
- **Step 4: Compute the Average**. Calculate the average value of $f(x)$ over all samples:

$$\text{Average} = \frac{1}{N}\sum_{i=1}^{N} f(x_i)$$

- **Step 5: Estimate the Integral**. Multiply the average by the interval length $(b-a)$ to estimate the integral:

$$\int_a^b f(x)\,dx \approx (b-a)\cdot\text{Average}$$

# Why Calculate the Average?

- **Intuition Behind the Average:**
  - To estimate the area under $f(x)$ over $[a, b]$, think of the average height of $f(x)$ across this interval.
  - The average of function values at sampled points gives a representative height:

$$\text{Average} = \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

  - This average height, when multiplied by the interval length $(b - a)$, provides an approximation of the area:

$$\text{Approximate Area} = (b - a) \times \text{Average}$$

  - If the samples are well-distributed, the average closely matches the true mean height of the function, leading to a good approximation of the integral.